



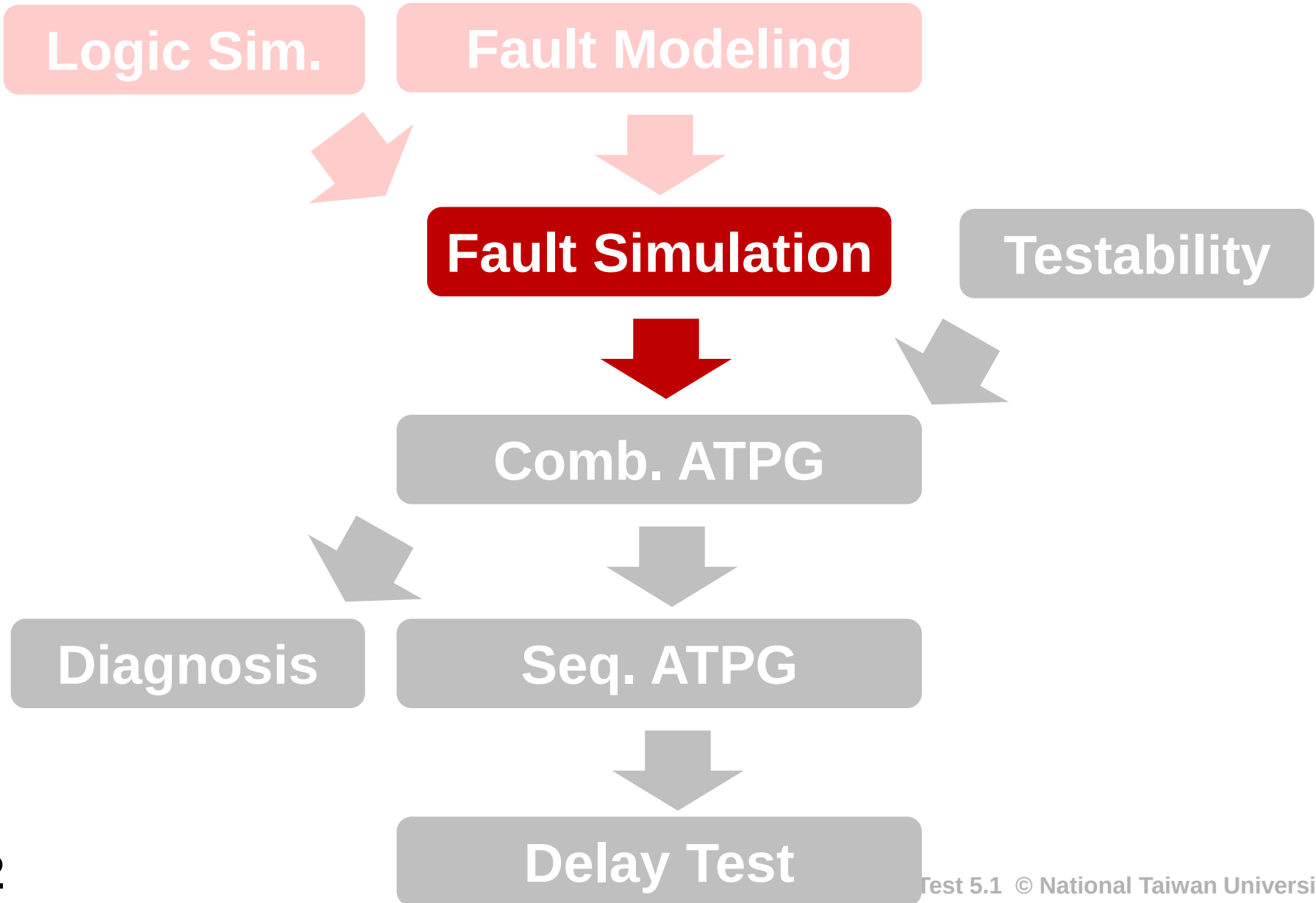
VLSI Testing

積體電路測試

Fault Simulation

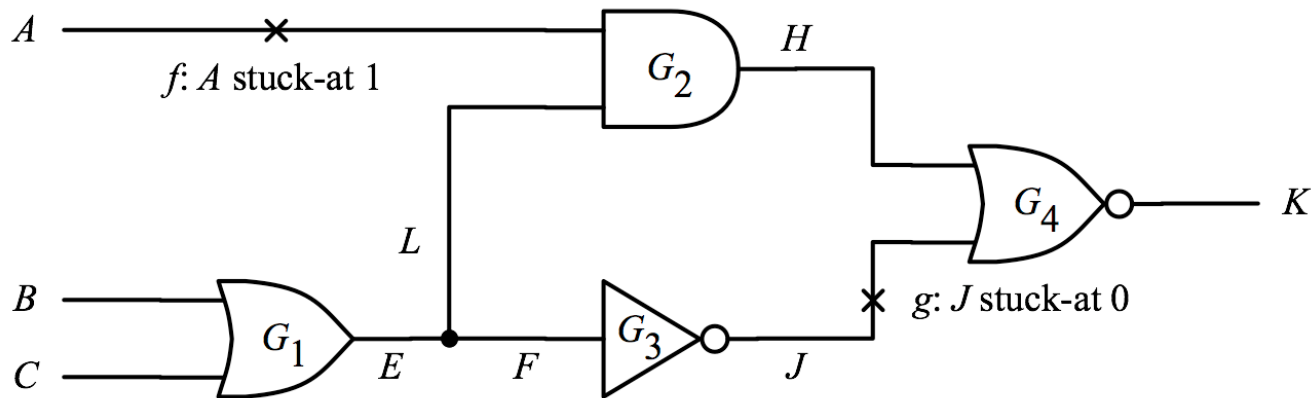
Professor James Chien-Mo Li 李建模
Lab. of Dependable Systems (LaDS)
Graduate Institute of Electronics Engineering
National Taiwan University

Course Roadmap (EDA Topics)



Motivating Problem

- Apply 3 test patterns: P_1, P_2, P_3 : $\{0,1,0\}$ $\{0,0,1\}$ $\{1,0,0\}$
 - Your manager asks you : **What is fault coverage?**



Pat.	Input			Internal					Output
	A	B	C	E	F	L	J	H	K
P_1	0	1	0	1	1	1	0	0	1
P_2	0	0	1	1	1	1	0	0	1
P_3	1	0	0	0	0	0	1	0	0

Why Am I Learning This?

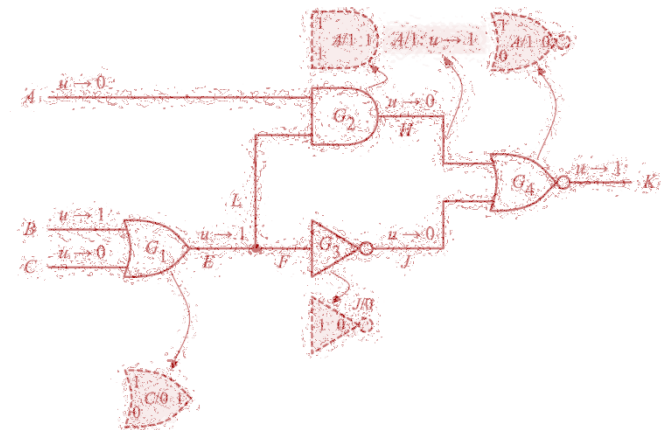
- Fault simulation can
 - ◆ 1. Determine **fault coverage**
 - ◆ 2. Guide **ATPG**
 - ◆ 3. **Diagnose** failed circuits

*“One sees qualities at a distance
and defects at close range.”*

(Victor Hugo)

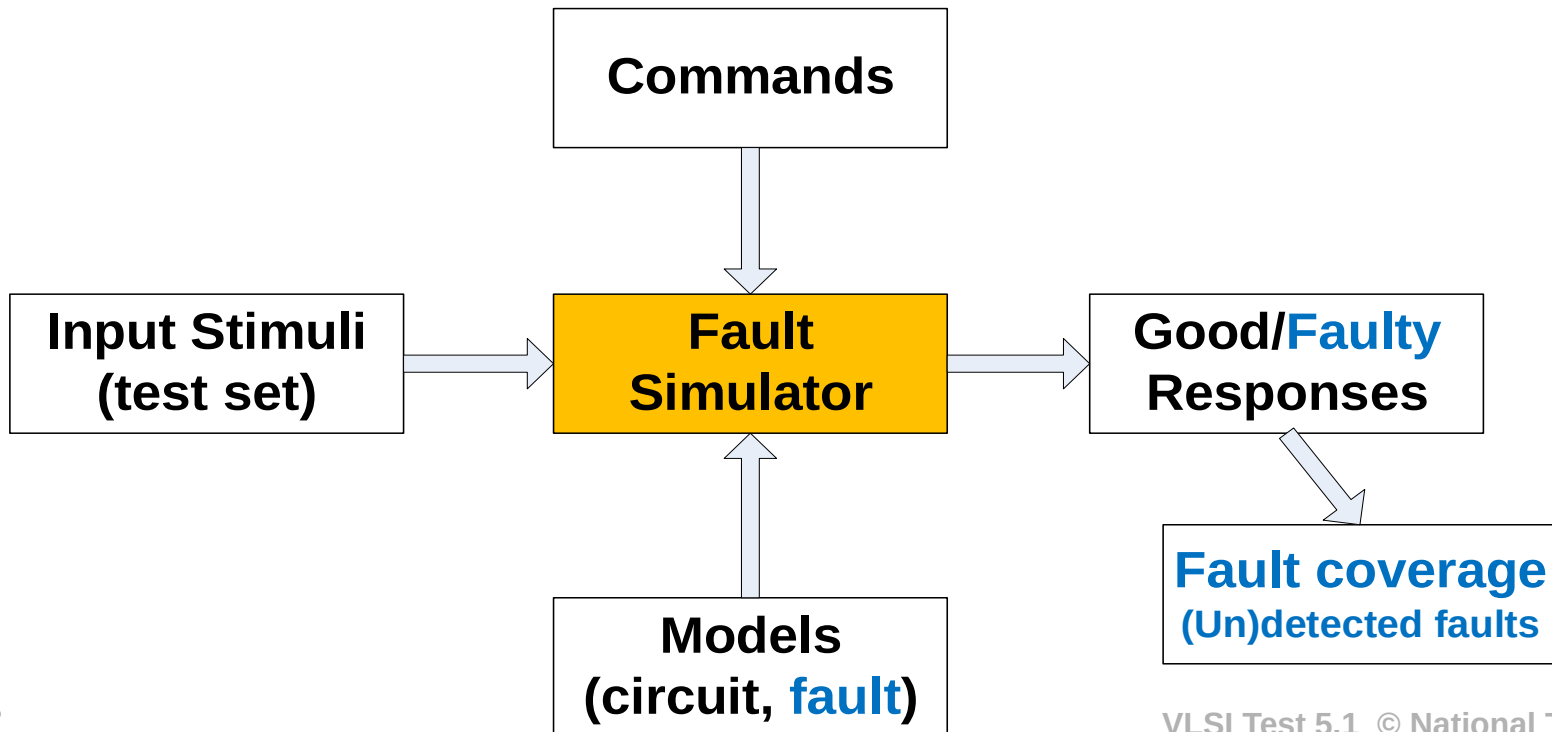
Fault Simulation

- Introduction
- Fault simulation techniques
- Comparison of fault simulation techniques
- Alternatives to fault simulation
- Issues of fault simulation
- Concluding remarks



What Is Fault Simulation?

- **Given:**
 - ♦ Circuit, fault model, test set
- **Determine:**
 - ♦ Output responses of faulty circuits
 - ♦ Detected faults, Undetected faults
 - ♦ Fault coverage

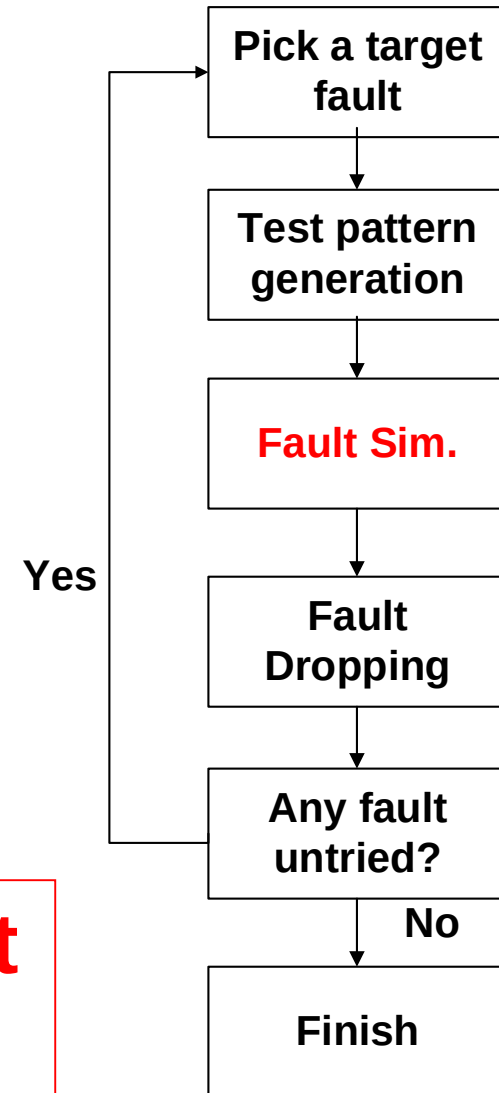


Applications of Fault Simulation

- 1. Evaluate quality of test sets (aka. **Fault Grading**)
 - ♦ Determine fault coverage of a test set
- 2. Automatic Test Pattern Generation (**ATPG**)
 - ♦ Identify detected faults and undetected faults
- 3. **Diagnosis** (See diagnosis chapter)
 - ♦ Generate *fault dictionary*
 - ♦ Find culprit fault responsible for failure
- Therefore, fault simulators can be used as either
 - ♦ Stand alone tool, or
 - * Fault grading
 - ♦ Embedded tool
 - * ATPG
 - * Diagnosis

Fault Sim. Embedded in ATPG

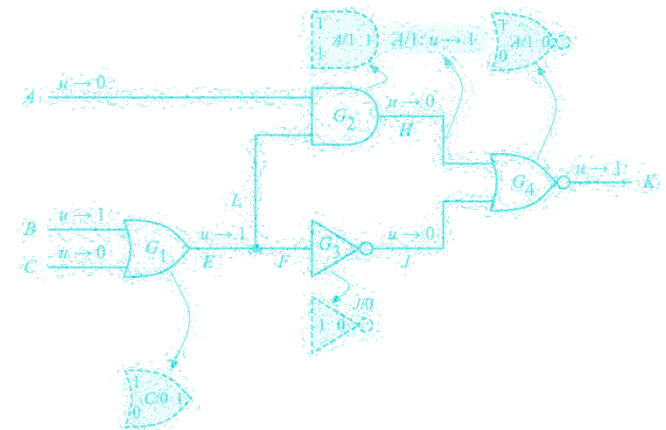
- After each test pattern generation
 - ♦ fault simulation
- **Fault dropping**
 - ♦ Remove detected faults from fault list
 - ♦ Prevent repeated test generation for detected faults
 - ♦ No dropping for diagnosis (see diagnosis chapter)



**Fault Sim. Very Important
for ATPG**

Fault Simulation

- Introduction
- Fault simulation techniques
 - ◆ Serial fault simulation
 - ◆ Parallel fault simulation (1965)
 - ◆ PPSFP (1985)
 - ◆ Deductive fault simulation (1972)
 - ◆ Concurrent fault simulation (1974)
 - ◆ Differential fault simulation (1989)
- Alternatives to fault simulation
- Issues of fault simulation
- Concluding remarks

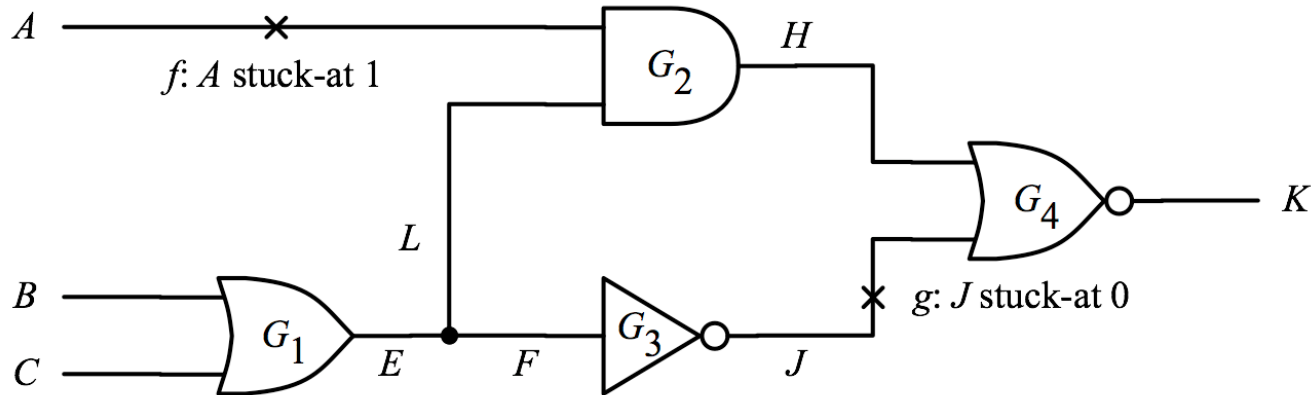


Serial Fault Simulation

- Simple Idea: Run a fault-free logic simulation, store good outputs
 - ◆ For every fault
 - * Modify good circuit (**fault injection**) to obtain a faulty circuit
 - * Run logic simulation on faulty circuit
 - * Compare faulty outputs with stored good outputs
 - fault is detected if they are **different**
- Advantages
 - ◆ Easy to implement (regular logic simulator).
 - ◆ Ability to simulate many fault models (stuck-at, delay, Br, ...)
- Disadvantage
 - ◆ Long CPU time

Example

- Consider only 2 faults: f , g Assume no fault dropping
- Given 3 patterns: P_1 , P_2 , P_3

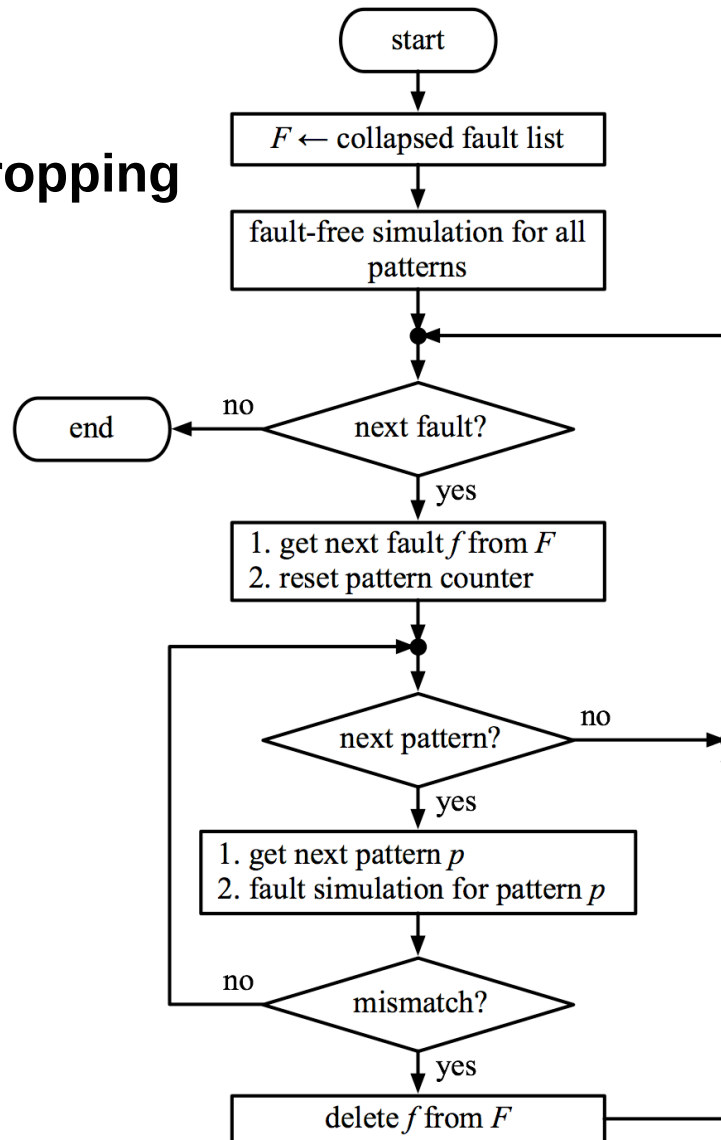


Pat. #	Input			Internal					Output		
	A	B	C	E	F	L	J	H	K_{good}	K_f	K_g
P_1	0	1	0	1	1	1	0	0	1	<u>0</u>	1
P_2	0	0	1	1	1	1	0	0	1	<u>0</u>	1
P_3	1	0	0	0	0	0	1	0	0	0	<u>1</u>

underline = detection

Serial Fault Simulation Flow

- Inner loop: patterns
- Outer loop: faults
 - ♦ Speedup with fault dropping

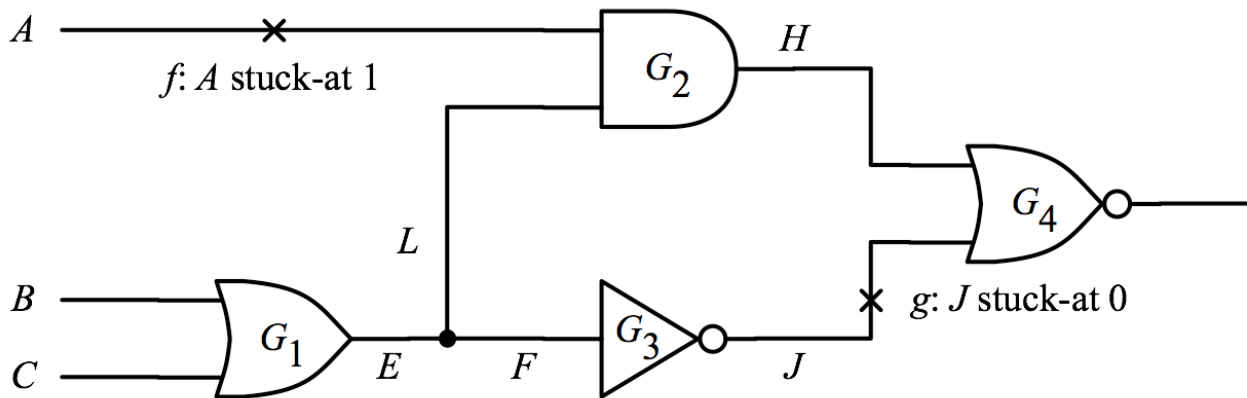


(WWW Fig 3.23)

Quiz

Q: Apply 3 test patterns: P_1, P_2, P_3 : $\{0,1,0\}$ $\{0,0,1\}$ $\{1,0,0\}$. Please use serial fault simulation to determine fault coverage =? Consider all 18 faults.

A:



Pat.	Input			Internal					Output
	A	B	C	E	F	L	J	H	K
P_1	0	1	0	1	1	1	0	0	1
P_2	0	0	1	1	1	1	0	0	1
P_3	1	0	0	0	0	0	1	0	0

fault	Det. by
A/1	P_1
A/0	-
B/1	
B/0	
C/1	
C/0	
E/1	
E/0	
F/1	
F/0	
H/1	
H/0	
J/1	
J/0	
K/1	
K/0	
L/1	
L/0	

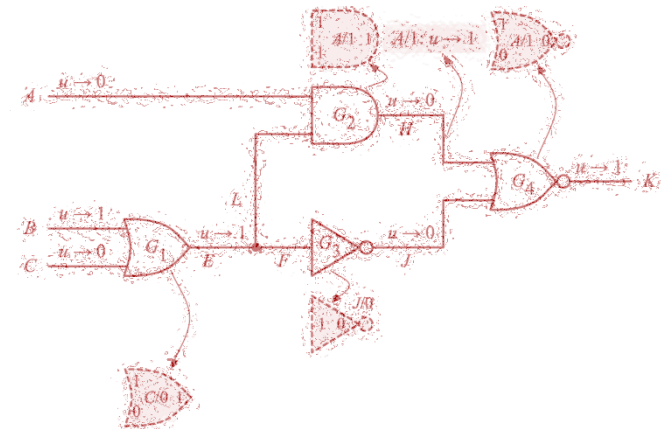
Complexity of Fault Simulation

- Fault Simulation Complexity $O(F \times P \times G)$
 - ♦ F : number of faults
 - ♦ P : number of test patterns
 - ♦ G : number of gates
- Comparison
 - Logic simulation: $O(G \times P)$
 - ♦ ATPG: $O(G \times 2^{\text{number_of_PI}} \times F)$
- So, use fault simulation to guide ATPG makes sense

ATPG > Fault Sim > Logic Sim

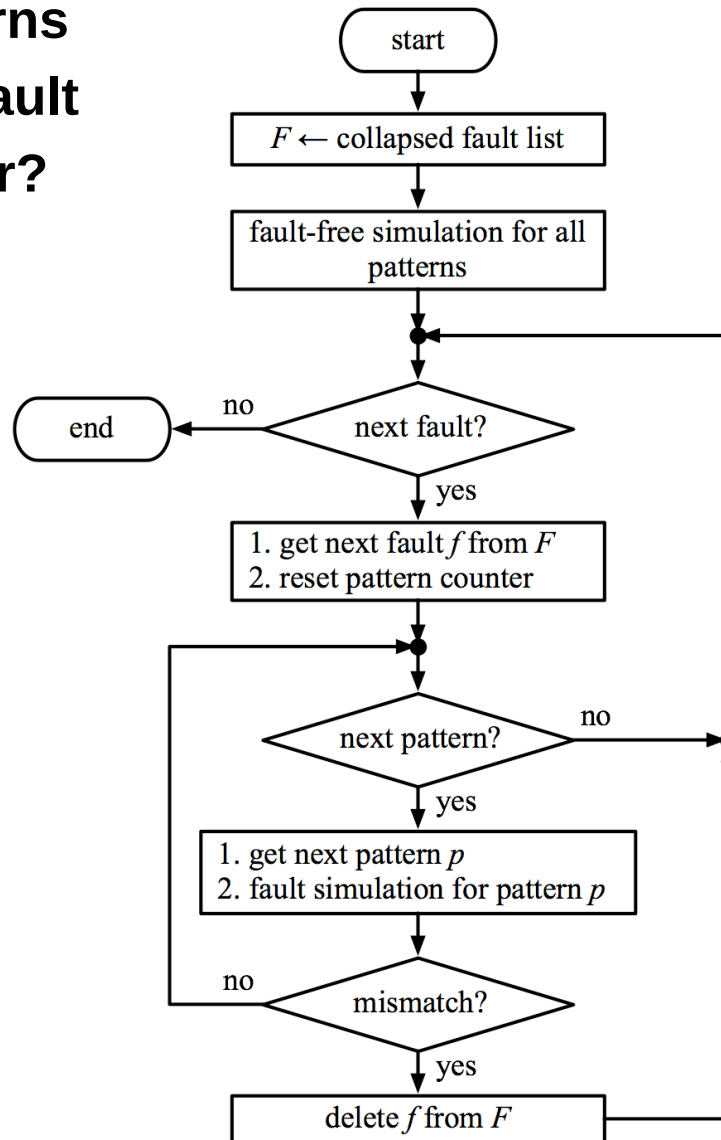
Fault Simulation

- Introduction
 - ◆ Fault simulation produces faulty circuit responses
 - ◆ Application: **fault grading, ATPG, diagnosis**
 - ◆ Fault simulation is polynomial time $O(P \times G \times F)$
- Serial fault simulation
 - ◆ Simulate one fault by one fault



FFT

- Q: why inner loop=patterns
 - ♦ why not inner loop=fault
 - ♦ So we can drop faster?



(WWW Fig 3.23)